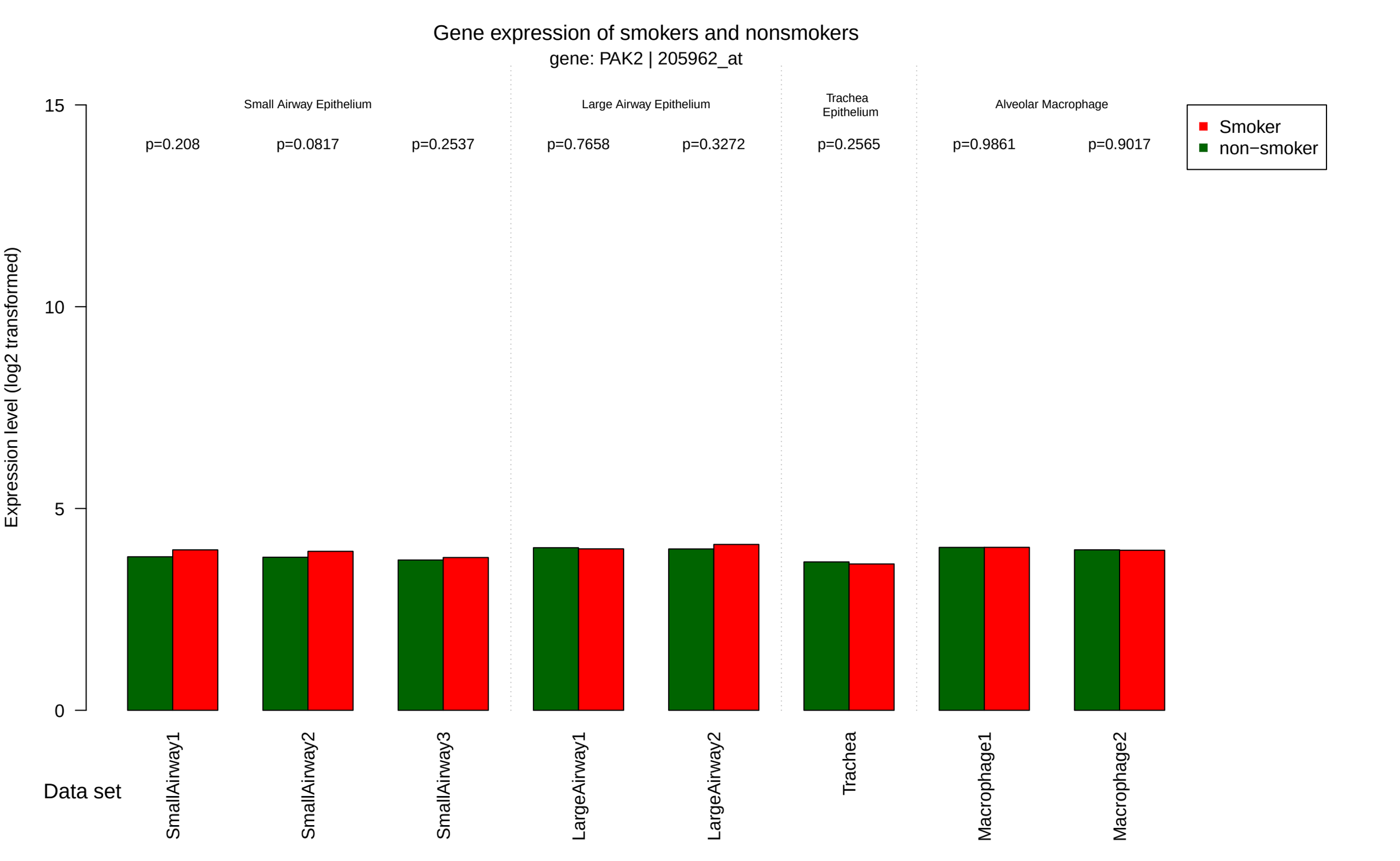
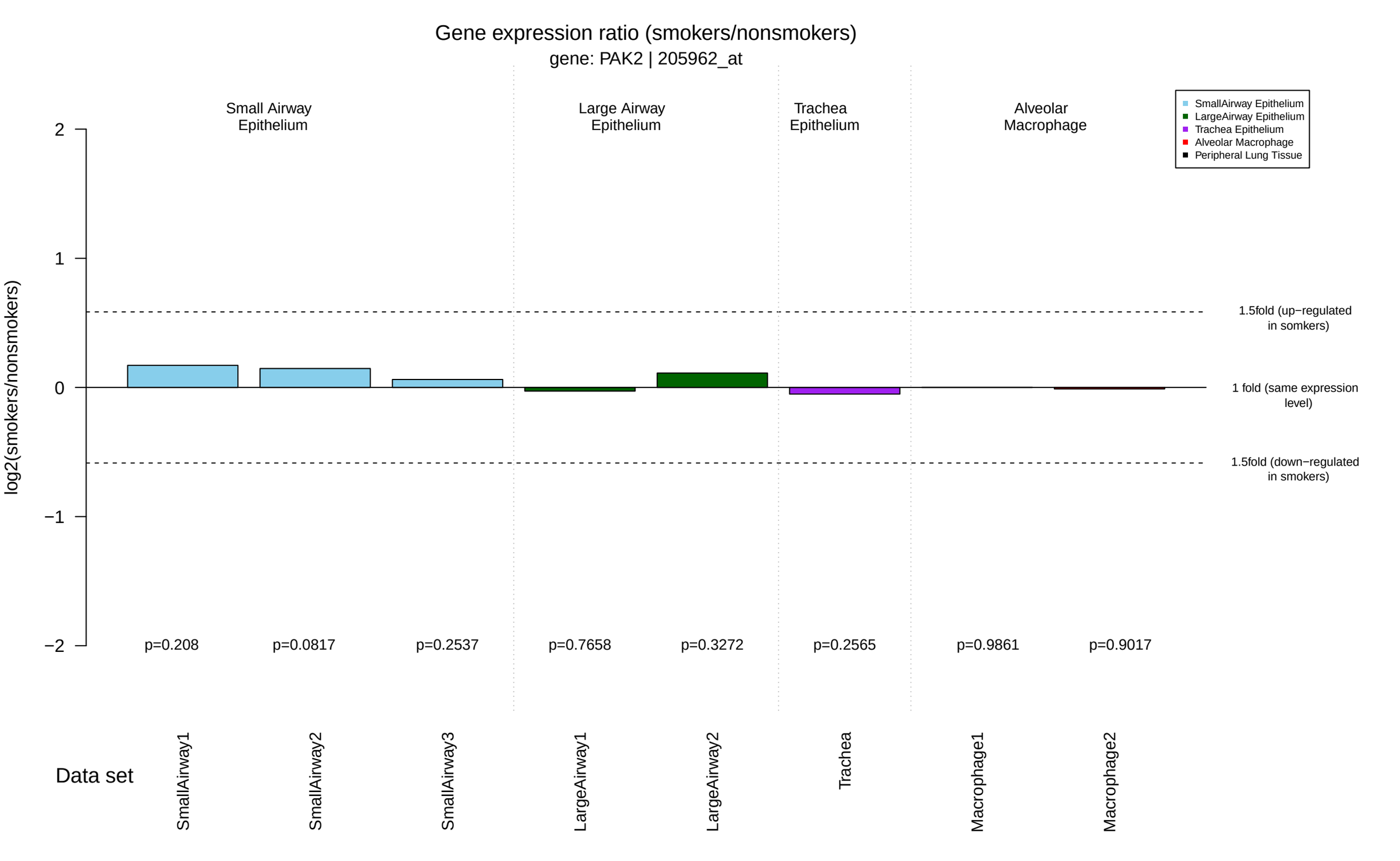
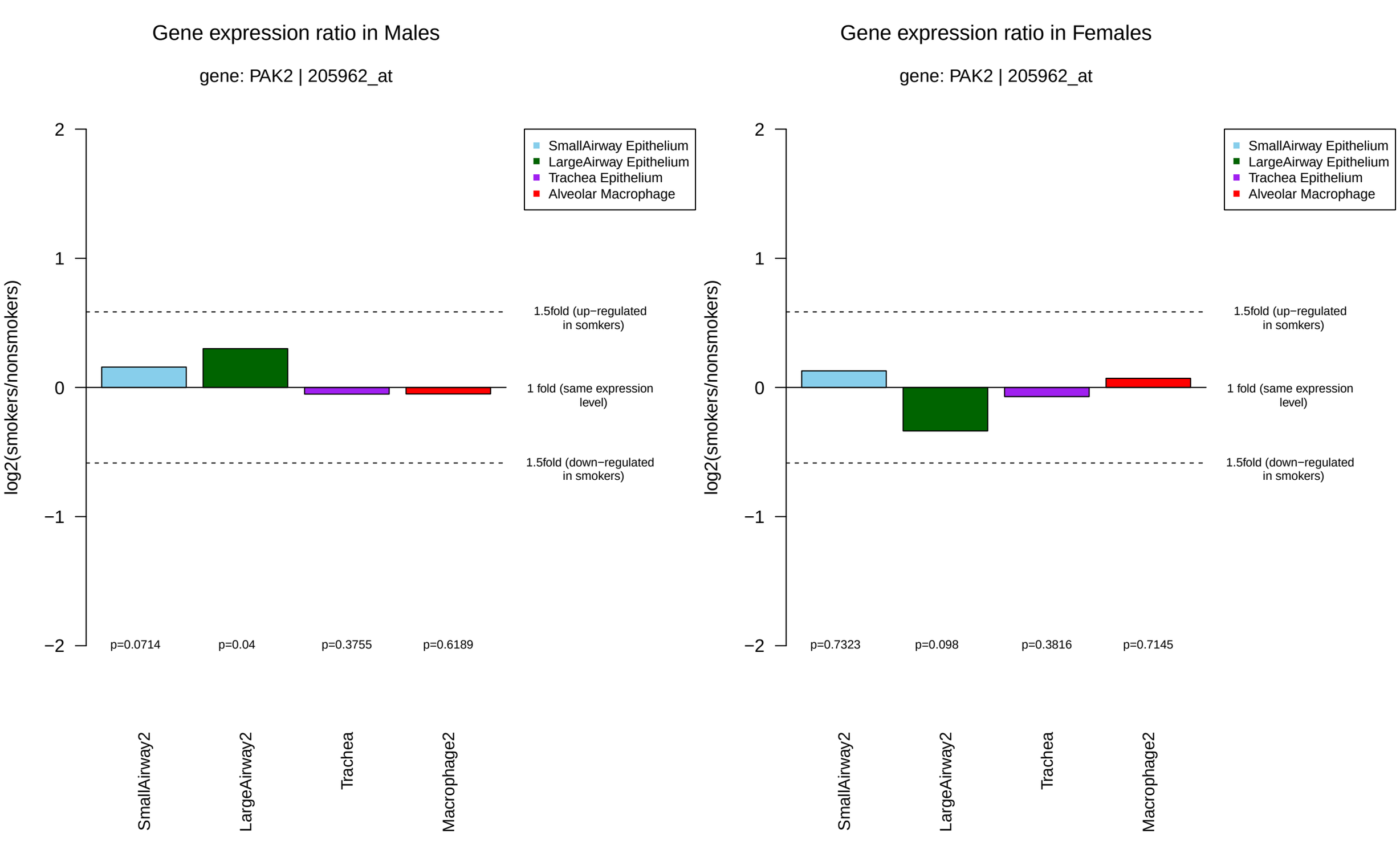
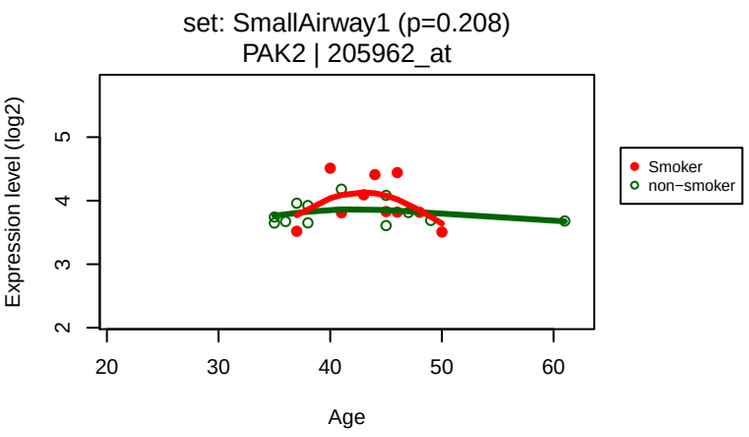
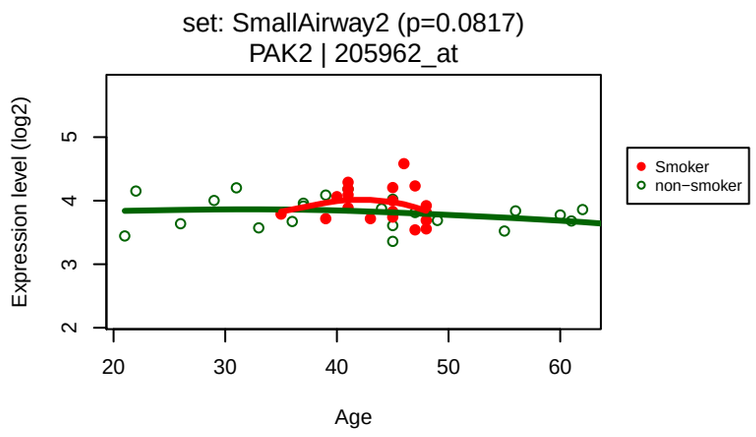
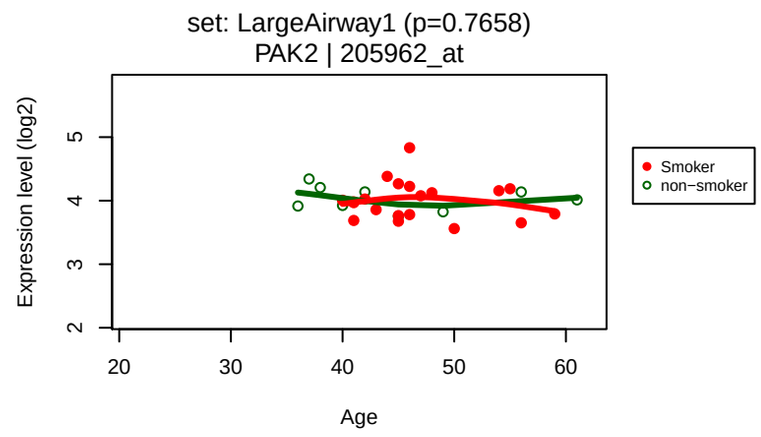

Gene expression of smokers and nonsmokers
gene: PAK2 | 205962_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.208
p=0.0817
p=0.2537
p=0.7658
p=0.3272
p=0.2565
p=0.9861
p=0.9017
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

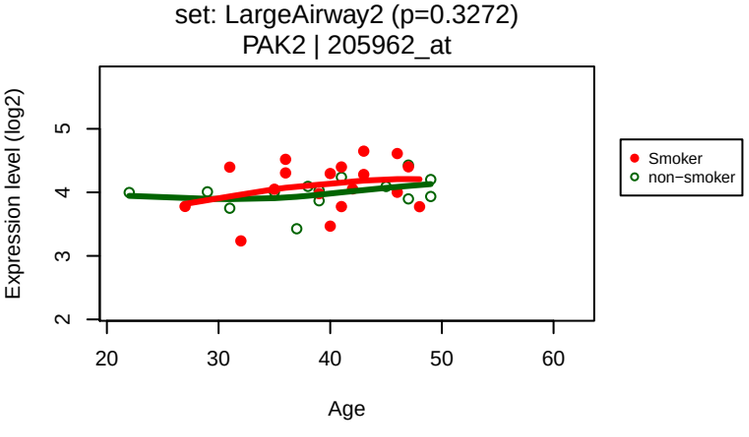

Gene expression ratio (smokers/nonsmokers)
gene: PAK2 | 205962_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.208
p=0.0817
p=0.2537
p=0.7658
p=0.3272
p=0.2565
p=0.9861
p=0.9017
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

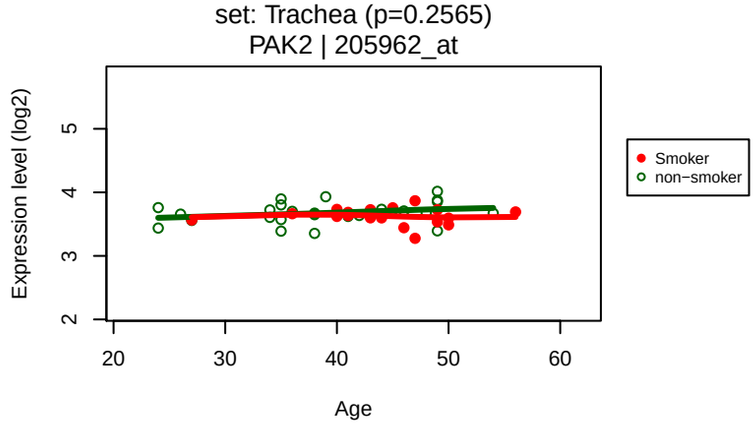

Gene expression ratio in Males
gene: PAK2 | 205962_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0714
p=0.04
p=0.3755
p=0.6189
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK2 | 205962_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7323
p=0.098
p=0.3816
p=0.7145
SmallAirway2
LargeAirway2
Trachea
Macrophage2

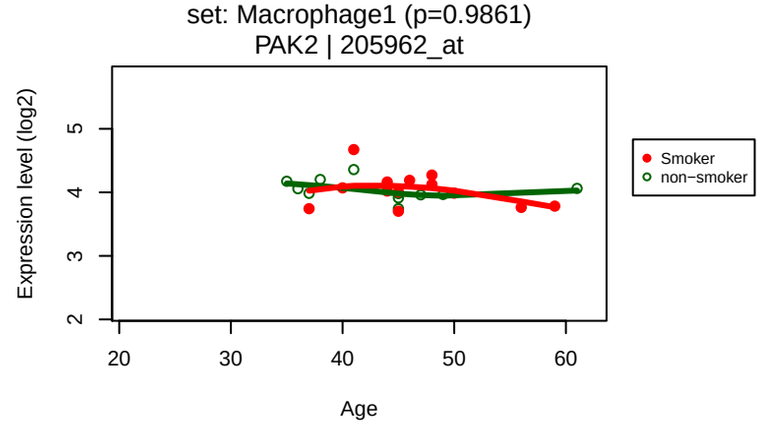

set: SmallAirway1 (p=0.208)
PAK2 | 205962_at
Expression level (log2)
Age
Smoker
non−smoker

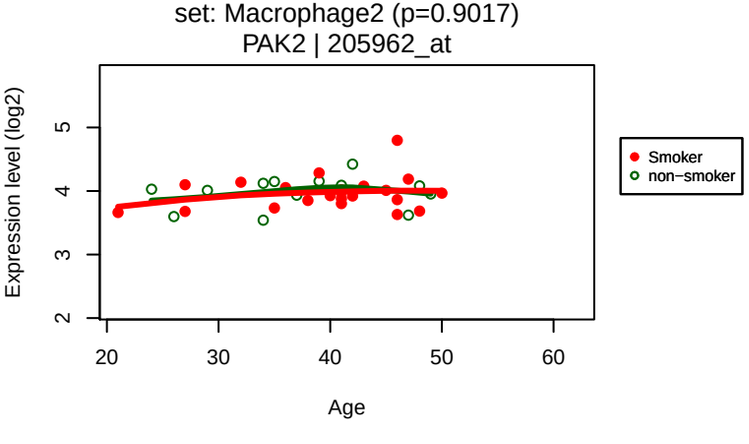

set: SmallAirway2 (p=0.0817)
PAK2 | 205962_at
Expression level (log2)
Age
Smoker
non−smoker

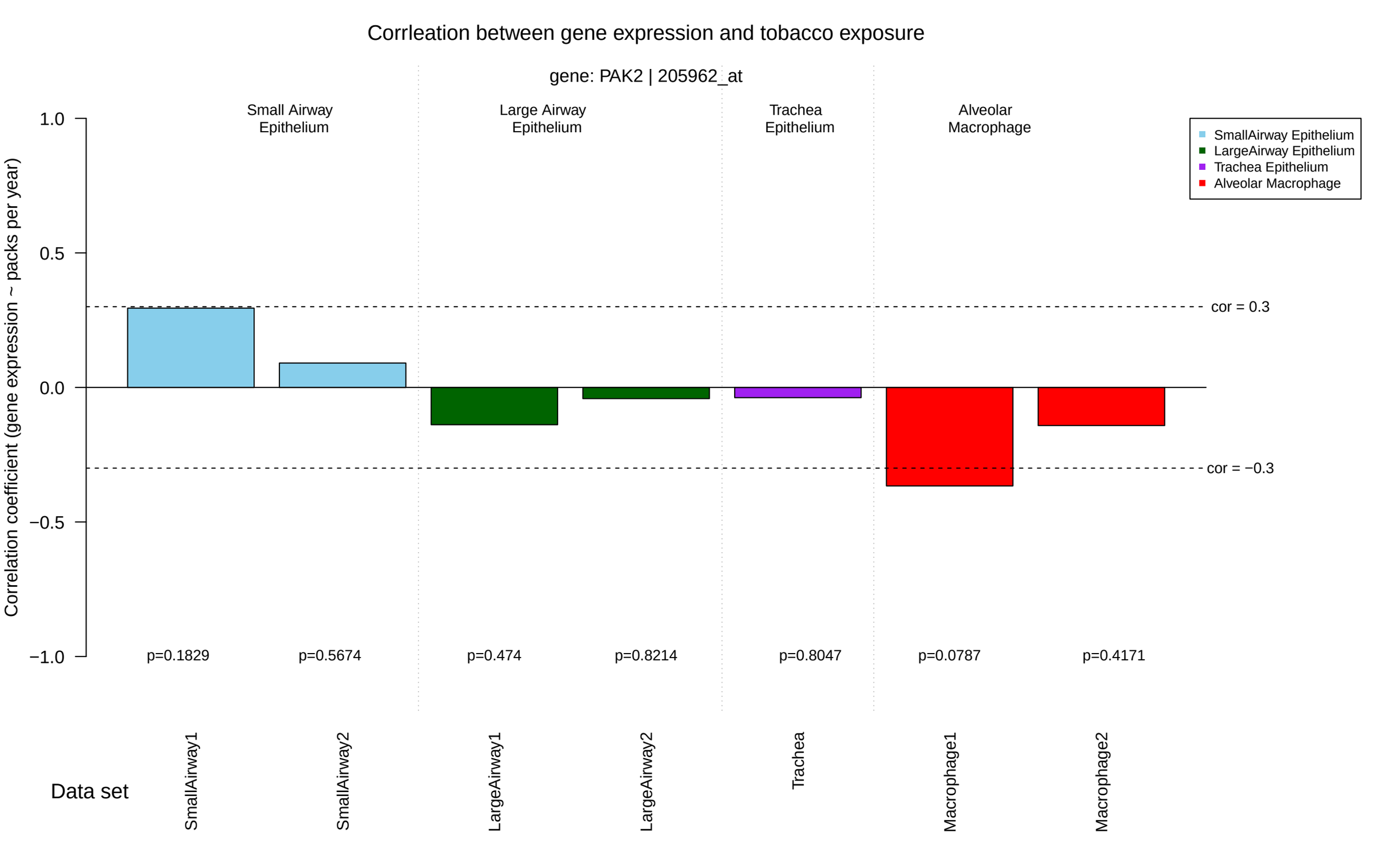

set: LargeAirway1 (p=0.7658)
PAK2 | 205962_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3272)
PAK2 | 205962_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2565)
PAK2 | 205962_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9861)
PAK2 | 205962_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9017)
PAK2 | 205962_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK2 | 205962_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1829
p=0.5674
p=0.474
p=0.8214
p=0.8047
p=0.0787
p=0.4171
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2